Rahul Mallick

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Thermodynamics Lab
Department of Mechanical Engineering
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Academic Status

Doctoral Research Scholar
· Thermodynamics Lab, Department of Mechanical Engineering, IIT Kharagpur 2024 – Present

Education

Ph.D. in Mechanical Engineering, Indian Institute of Technology (IIT) Kharagpur 2024 – Present
· Advisor: Prof. Aditya Bandopadhyay
· Focus: Interfacial Electrohydrodynamics, Boundary Element Method (BEM) for Deformable Droplets, Multiphase Lattice Boltzmann Method (LBM), and Electrokinetics.

Master of Engineering (ME) in Mechanical Engineering, Jadavpur University May 2024
· First Class (Specialization: Thermal)

Bachelor of Technology (B.Tech) in Mechanical Engineering, Aliah University May 2021
· First Class Honors

Research Interests

Primary: Interfacial Electrohydrodynamics (EHD), Multiphase Flows, Microfluidics
Secondary: Boundary Element Method (BEM), Lattice Boltzmann Method (LBM), High-Performance Scientific Computing

Publications

1. **Rahul Mallick** and Aditya Bandopadhyay, "Oscillatory Electrohydrodynamics of a Droplet in an Alternating Quadrupole Electric Field: Boundary Element Method Analysis," *International Journal of Multiphase Flow* (*Submitted / Under Review*), 2026.

Conference Presentations

1. Presentations in progress.

Honors, Fellowships, and Awards

Institute Doctoral Research Fellowship
· Ministry of Education (MoE), Government of India & IIT Kharagpur 2024 – Present

Teaching Experience

Indian Institute of Technology Kharagpur
· Experimental Techniques in Thermal Engineering (ME60227)
· Applied Thermodynamics Lab (ME3022)

Skills

Numerical Methods: Boundary Element Method (BEM), Lattice Boltzmann Method (LBM)

Programming & HPC: C++, Modern Fortran, Python (NumPy, SciPy, Matplotlib), CUDA GPU Acceleration, OpenMP, MPI, CMake, Git

CFD & Visualization: ParaView (Python scripting), OpenFOAM, COMSOL Multiphysics, LaTeX

Languages: English, Bengali, Hindi

References

Prof. Aditya Bandopadhyay (Ph.D. Advisor)
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Indian Institute of Technology Kharagpur
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Doctoral Scrutiny Committee (DSC)
Department of Mechanical Engineering
Indian Institute of Technology Kharagpur
Kharagpur, West Bengal 721302, India